Ethics, employment and health and safety

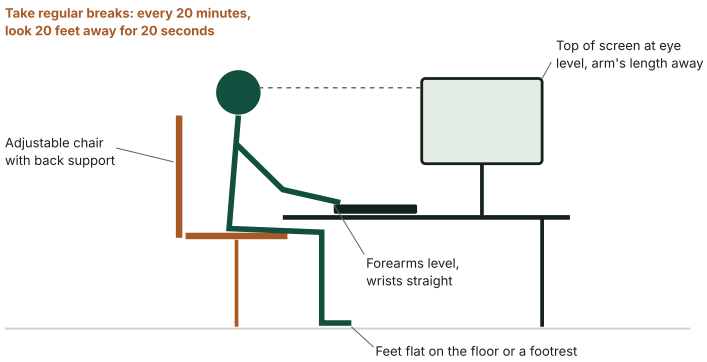
L23
FACT FILE

Specification: students should be able to

- Describe the ethical impact of technology on society, referring to internet misuse, access to personal information, social media misuse, the implications of GPS and tracking, and concerns about the security of personal data
- Describe the impact of digital technology on employment, including increased job opportunities in the digital technology and computing sector, job displacement, changes in work patterns and the need for upskilling
- Demonstrate understanding of digital technology related health and safety issues, including RSI, back strain and eye strain
- Identify the measures that both the employee and employer should take to promote good health and safety practice in the workplace, including the Health and Safety (Display Screen Equipment) Regulations 1992 (in Northern Ireland, the Health and Safety (Display Screen Equipment) Regulations (Northern Ireland) 1992)

CCEA fact file: Moral and Ethical, Fact File 20: Changes in employment, Health and Safety

Key words: ethics, internet misuse, acceptable use policy, phishing, social media misuse, cyberbullying, trolling, GPS, tracking, privacy, job displacement, upskilling, teleworking, RSI, back strain, eye strain, ergonomic, Display Screen Equipment Regulations (NI) 1992



A correctly set-up computer workstation

Technology and society

Ethics is about what is right and wrong, not just what is legal. Digital technology brings huge benefits, but it also raises ethical questions. The internet can be misused for crimes such as phishing, stealing account details and piracy, and for spreading fake news or hateful content. Huge amounts of personal information are collected through loyalty cards, card payments, cookies, phone location and CCTV, often without people really noticing. Social media can be misused for cyberbullying, trolling, fake profiles and sharing images without consent. GPS and tracking help with navigation and safety, but can invade privacy. Technology also changes jobs: it creates new ones in the digital sector, replaces others, changes when and where people work, and means everyone needs to keep learning. Finally, spending long hours at a computer can cause health problems such as RSI, back strain and eye strain, so employers and employees both have responsibilities to work safely.

Misuse, privacy and tracking

Internet misuse includes crimes and behaviour that harms others or wastes time. Phishing uses fake but official-looking messages to trick people into giving details, and stolen passwords are then used to buy goods or empty accounts. Plagiarism and illegal downloading take credit or profit from other people's work. Employers deal with staff misuse through an acceptable use policy. On social media the main misuses are cyberbullying, trolling, hate speech, fake profiles used for identity theft and oversharing personal details. Mark schemes treat trolling and cyberbullying as different points: bullying targets one person repeatedly, while trolling aims to provoke anyone. GPS uses satellites to work out a device's exact position. It helps parents, delivery firms, emergency services and carers of people with dementia, but tracking without consent invades privacy and could allow stalking. When data is stored in the cloud on servers abroad, it may be protected by weaker laws, which raises concerns about its security.

Jobs and work patterns

Digital technology has created many new jobs, such as software engineers, web and app designers, game developers, cyber security and digital forensics analysts, and database administrators. Northern Ireland has a growing technology sector, especially around Belfast. At the same time, job displacement happens when technology takes over work people used to do: robots assemble and paint cars, self-service tills replace checkout staff, and online banking has led to branch closures. Work patterns have changed too. Teleworking means using ICT, such as laptops, broadband, cloud apps and video calls, to work from home. It saves travel time and reduces pollution, but it can blur the line between home and work. Global companies and 24-hour call centres mean more shift work and staff spread across time zones. Because technology changes quickly, workers need upskilling, which is training in new skills. Good employers provide training because it improves productivity, staff morale and staff retention.

Staying healthy at a computer

Repetitive strain injury (RSI) is caused by repeating the same small movements, such as typing or clicking, for long periods. It causes pain and stiffness in the hands, wrists, arms or shoulders. Back strain comes from poor posture and unsuitable chairs. Eye strain is caused by staring at a screen for too long, especially with glare or poor lighting, and leads to headaches and blurred vision. Employees should take regular breaks, sit correctly, adjust their chair and screen and use wrist rests. Employers must provide a safe working environment under the Health and Safety at Work (Northern Ireland) Order 1978. Under the Health and Safety (Display Screen Equipment) Regulations (Northern Ireland) 1992, they must assess workstations, plan breaks or changes of activity, train staff and provide eye tests for regular screen users. They should supply adjustable chairs, footrests, ergonomic keyboards and mice, anti-glare screens and good lighting. Other safety measures include tidy cables, PAT testing of equipment, keeping food and drink away from computers and having CO_2 fire extinguishers.

Key terms from the mark schemes

Phishing
Using fake emails, texts or adverts that look official to trick people into giving away personal details such as passwords.

Acceptable use policy
Rules that employees or pupils agree to that set out how they may use an organisation's computers and internet.
`Credited: AUP`

Trolling
Deliberately posting provocative or offensive messages online to upset people or start arguments; credited separately from cyberbullying.

Fake news
Deliberately false or misleading information that is spread online, often made to look like real news.

Job displacement
When jobs are lost or changed because digital technology now does work that people used to do.

Upskilling
Training workers in new skills, especially digital skills, so they can keep up with changes in technology.
`Credited: retraining`

Repetitive strain injury (RSI)
Pain or injury in the hands, wrists, arms or shoulders caused by repeating the same movements, such as typing, for long periods.
`Credited: RSI`

Cyberbullying
Using technology, such as social media or messages, to repeatedly hurt, threaten or embarrass a particular person.

GPS
Global Positioning System: uses signals from satellites to work out the exact location of a device in real time.
`Credited: Global Positioning System`

Automation
Machines or software carrying out tasks that were previously done by people.

Teleworking
Using ICT, such as broadband, laptops and video conferencing, to work from home instead of at the workplace.
`Credited: remote working / working from home`

Ergonomic
Designed to fit the user and reduce physical strain, e.g. adjustable chairs, ergonomic keyboards and wrist rests.

Exam focus

This lesson often provides the extended-writing questions, which are marked for quality of written communication. 2021 Q8(a) asked about social media misuse and 2023 Q11(a) about the impact on employment, both for 6 marks. For these, cover several different points, explain each with an example, and write in organised paragraphs. Health and safety appeared in 2022 Q9(a) (RSI and monitor set-up), 2024 Q9(b) (health risks) and 2025 Q10, where examiners found answers about posture and ergonomics were often poorly targeted. Read whether the question asks about the employee or the employer and answer for that person: 'take a break' is an employee measure; 'provide adjustable chairs' is an employer measure. Match the measure to the problem asked about, so a wrist rest helps RSI, not eye strain. 2024 Q11(b) covered online harms, and 2019 Q1(f)–(g) and the key-term matching questions (2021 Q11, 2024 Q10) tested eye strain, GPS and job displacement.

`Unit 1 2025 Q10`	`[4]`	RSI/posture causes and ergonomic prevention
`Unit 1 2024 Q11`	`[6]`	Legislation matching, online harms, protection methods
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q9`	`[7]`	Input devices, health risks, gaming and internet risks
`Unit 1 2023 Q11(a)`	`[6]`	Impacts on employment (QWC)

Check yourself

1. Describe two ethical concerns about the use of GPS tracking. [4]
2. Ballymena Academy's ICT co-ordinator is writing a talk for Year 10 about how the internet is misused. Describe three different examples of misuse that the talk could include. [6]
3. Describe two ways in which a supermarket can collect personal data about its customers without them fully realising it. [4]

Answers: 1. Two concerns, each identified (1) and described (1): a person's location can be tracked without their consent (1) which invades their privacy (1); could be used for stalking or controlling someone (1) by following their live location (1); employers may monitor staff constantly (1), even outside working hours (1); apps may sell location data to advertisers (1) without users realising (1); geotagged posts can reveal home or school (1), putting people at risk (1). 2. Three ways, each identified (1) and described (1): phishing (1) – fake official-looking emails/adverts trick people into giving personal details (1); theft of usernames and passwords (1) – used to access accounts and make purchases (1); piracy (1) – illegally downloading copyrighted music, films or software (1); plagiarism (1) – copying work and claiming it as your own (1); spreading fake news or hateful/violent content (1) – misleads or harms people (1); using the dark web to access illegal material (1); employees using work internet for personal use (1) – wastes time, leading to acceptable use policies (1). 3. Two ways, each identified (1) and described (1): loyalty cards (1) record what each customer buys and when, used for targeted advertising (1); card payment records (1) show shopping patterns, which may be shared or sold (1); online shopping accounts/cookies (1) track browsing and purchases (1); CCTV (1) records customers in store (1); app/Wi-Fi location data (1) shows when customers visit (1).